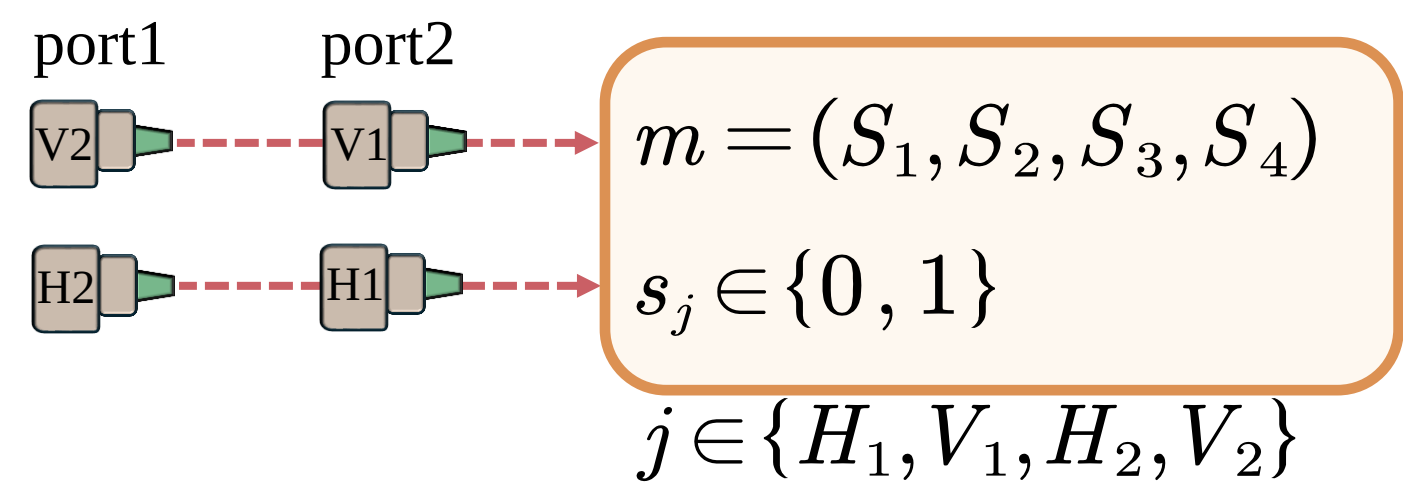


测量记录到输入端效应算符

$$E_m^{\text{in}} = \Phi^\dagger(F_m), \quad \Phi = \mathcal{M}_{\text{det}} \circ \text{Ad}_{U_{BS}} \circ \mathcal{L} \circ \mathcal{V}$$

探测端记录效应



Heralded success

$$E_j^{(1)} = I - (1 - d_j)(I - \eta_j n_j)$$

$$E_j^{(0)} = I - E_j^{(1)}$$

η_j : 探测效率, d_j : 暗记数概率, n_j : 光子数算符

Joint record

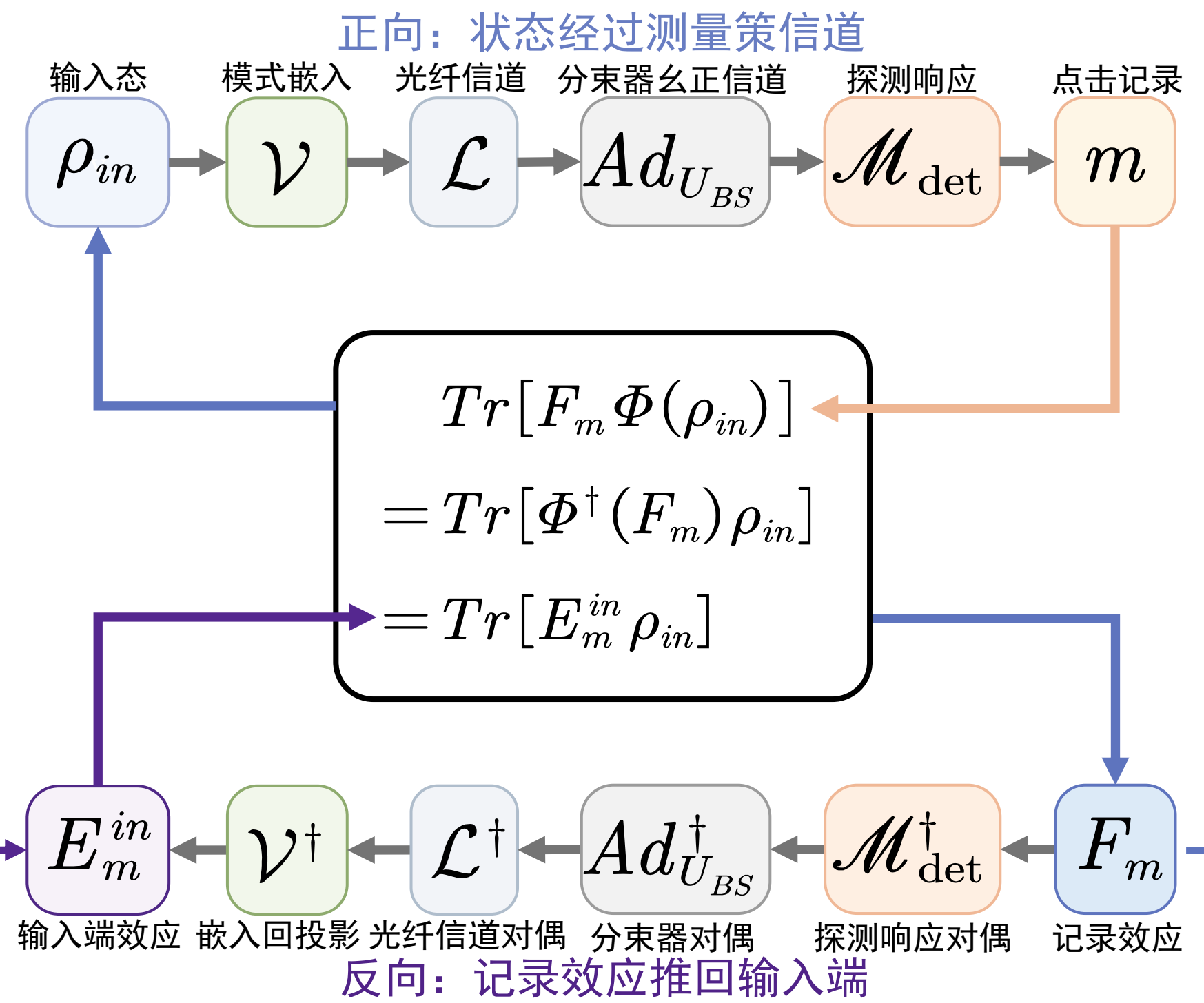
$$F_m = \bigotimes_{j=1}^4 E_j^{(s_j)}$$

点击模式m在探测端定义, 随后整体推回输入端

$$S = \{H_1 V_2, V_1 H_2, H_1 V_1, H_2 V_2, \dots\}$$

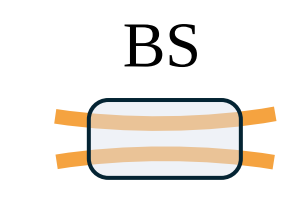
成功记录集合示例

概率等效的两种计算方向



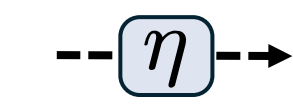
把测量侧器件移到算符一侧后, 状态侧只与 E_m^{in} 收缩。

模块对偶算符表达式



分束器

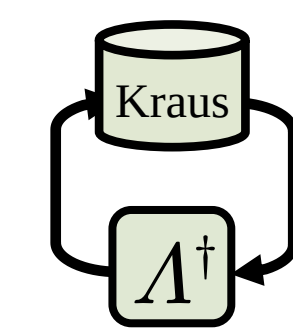
$$\text{Ad}_{U_{BS}}^\dagger(E) = U_{BS}^\dagger E U_{BS}$$



损耗/偏振信道

$$\mathcal{L}^\dagger(E) = \sum_{\alpha} K_{\alpha}^\dagger E K_{\alpha}$$

Kraus对偶作用



模式嵌入

$$\mathcal{V}^\dagger(E) = \mathcal{V}^\dagger E \mathcal{V}$$

$$\mathcal{V}^\dagger \mathcal{V} = I_m$$

输入端效应 (整体合成)

$$E_m^{\text{in}} = \mathcal{V}^\dagger \mathcal{L}^\dagger (U_{BS}^\dagger \mathcal{M}_{\text{det}}^\dagger (F_m) U_{BS}) \mathcal{V}$$

记录集合线性求和

$$p_S = \sum_{m \in S} \text{Tr}(E_m^{\text{in}} \rho_{in})$$

$$F_m = F_m^{\text{true}} + F_m^{\text{bg}} + F_m^{\text{dark}}$$
$$\Rightarrow E_m^{(\cdot)} = \Phi_m^\dagger F_m^{(\cdot)}$$

成功模式、真成功、背景辅助的暗记数辅助均可在效应层线性处理并求和。